

Power BI Data Model

[movie_rental_analysis.pbix](#)

Business tables: 16 | Relationships: 16 | Internal auto-date: 20

How to read

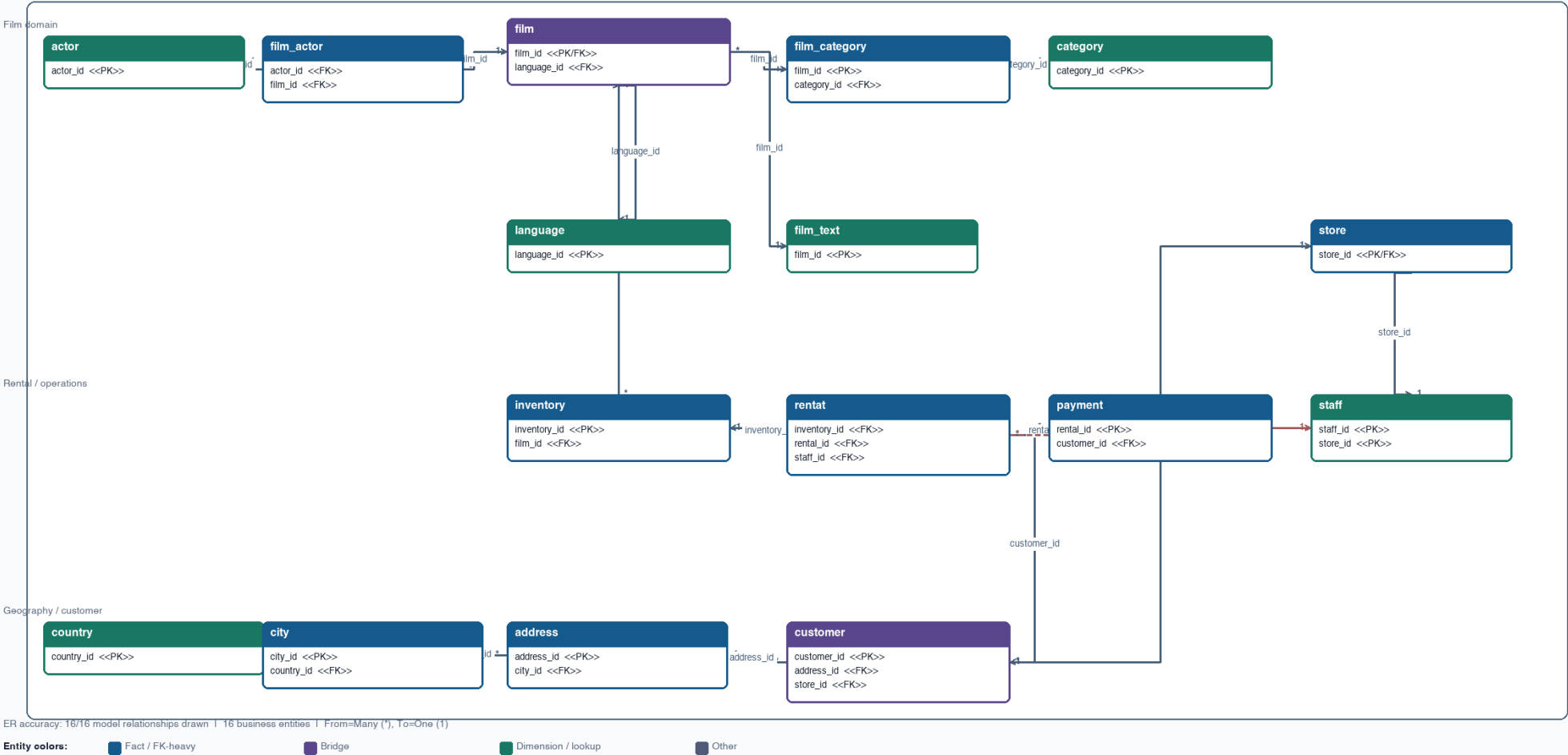
1. Next page(s): visual ER diagram (tables + join lines). Zoom in your PDF reader.
2. * = many / FK side (Power BI From). 1 = one / PK side (To).
3. After the ER diagram: every relationship printed in full (nothing clipped).
4. Later pages list every business column (PK/FK marked) and internal date tables.

ER diagram - movie_rental_analysis.pbix

ER diagram
movie_rental_analysis.pbix

ER diagram (business tables)

Exact joins from the PBIX model. * = many (FK side) 1 = one (lookup / PK side) Dashed red = inactive relationship
Layout: domain swimlanes (Film / Operations / Geography) - matches this PBIX



ER relationships checklist (16)

Many side (FK) -> One side (PK) | Card / Filter / Active

1. address.city_id -> city.city_id

[*] many = address [1] one = city | M:1 | filter: Both | ACTIVE

2. city.country_id -> country.country_id

[*] many = city [1] one = country | M:1 | filter: Single | ACTIVE

3. customer.address_id -> address.address_id

[*] many = customer [1] one = address | M:1 | filter: Both | ACTIVE

4. customer.store_id -> store.store_id

[*] many = customer [1] one = store | M:1 | filter: Single | ACTIVE

5. film.film_id -> film_category.film_id

[*] many = film [1] one = film_category | M:1 | filter: Single | ACTIVE

6. film.film_id -> film_text.film_id

[*] many = film [1] one = film_text | M:1 | filter: Single | ACTIVE

7. film.language_id -> language.language_id

[*] many = film [1] one = language | M:1 | filter: Single | ACTIVE

8. film_actor.actor_id -> actor.actor_id

[*] many = film_actor [1] one = actor | M:1 | filter: Single | ACTIVE

9. film_actor.film_id -> film.film_id

[*] many = film_actor [1] one = film | M:1 | filter: Both | ACTIVE

10. film_category.category_id -> category.category_id

[*] many = film_category [1] one = category | M:1 | filter: Single | ACTIVE

11. inventory.film_id -> film.film_id

[*] many = inventory [1] one = film | M:1 | filter: Both | ACTIVE

12. payment.customer_id -> customer.customer_id

[*] many = payment [1] one = customer | M:1 | filter: Both | ACTIVE

13. **rentat.inventory_id -> inventory.inventory_id**

[*] many = rentat [1] one = inventory | M:1 | filter: Both | ACTIVE

14. **rentat.rental_id -> payment.rental_id**

[*] many = rentat [1] one = payment | M:1 | filter: Both | ACTIVE

15. **store.store_id -> staff.store_id**

[*] many = store [1] one = staff | M:1 | filter: Single | ACTIVE

16. **rentat.staff_id -> staff.staff_id**

[*] many = rentat [1] one = staff | M:1 | filter: Single | INACTIVE

Schema layers (left = lookup, right = FK side)

Layer 0: actor (dim), category (dim), country (dim), film_text (dim), language (dim), staff (dim)

Layer 1: city (fact), film_category (fact), store (fact)

Layer 2: address (fact), film (bridge)

Layer 3: customer (bridge), film_actor (fact), inventory (fact)

Layer 4: payment (fact)

Layer 5: rentat (fact)

Business tables (all columns)

PK / FK marks come from model relationships and key flags.

actor (dim, 4 cols)

- actor_id [PK] (Int64)
- first_name (string)
- last_name (string)
- last_update (datetime64[ns])

address (fact, 9 cols)

- address_id [PK] (Int64)
- address (string)
- address2 (string)
- district (string)
- city_id [FK] (Int64)
- postal_code (Int64)
- phone (Int64)
- location (string)
- last_update (datetime64[ns])

category (dim, 3 cols)

- category_id [PK] (Int64)
- name (string)
- last_update (datetime64[ns])

city (fact, 4 cols)

- city_id [PK] (Int64)
- city (string)
- country_id [FK] (Int64)
- last_update (datetime64[ns])

country (dim, 3 cols)

- country_id [PK] (Int64)
- country (string)
- last_update (datetime64[ns])

customer (bridge, 10 cols)

- customer_id [PK] (Int64)
- store_id [FK] (Int64)
- first_name (string)
- last_name (string)
- email (string)
- address_id [FK] (Int64)
- active (Int64)
- create_date (datetime64[ns])
- last_update (datetime64[ns])
- Employment Duration [CALC] (Int64)

film (bridge, 13 cols)

- film_id [PK,FK] (Int64)
- title (string)
- description (string)
- release_year (Int64)
- language_id [FK] (Int64)
- original_language_id (Int64)
- rental_duration (Int64)
- rental_rate (Float64)
- length (Int64)
- replacement_cost (Float64)
- rating (string)
- special_features (string)
- last_update (datetime64[ns])

film_actor (fact, 3 cols)

- actor_id [FK] (Int64)
- film_id [FK] (Int64)
- last_update (datetime64[ns])

film_category (fact, 3 cols)

- film_id [PK] (Int64)
- category_id [FK] (Int64)
- last_update (datetime64[ns])

film_text (dim, 3 cols)

- film_id [PK] (Int64)
- title (string)
- description (string)

inventory (fact, 4 cols)

- inventory_id [PK] (Int64)
- film_id [FK] (Int64)
- store_id (Int64)
- last_update (datetime64[ns])

language (dim, 3 cols)

- language_id [PK] (Int64)
- name (string)
- last_update (datetime64[ns])

payment (fact, 7 cols)

- payment_id (Int64)
- customer_id [FK] (Int64)
- staff_id (Int64)
- rental_id [PK] (Int64)
- amount (Float64)

- payment_date (datetime64[ns])
- last_update (datetime64[ns])

rentat (fact, 7 cols)

- rental_id [FK] (Int64)
- rental_date (datetime64[ns])
- inventory_id [FK] (Int64)
- customer_id (Int64)
- return_date (datetime64[ns])
- staff_id [FK] (Int64)
- last_update (datetime64[ns])

staff (dim, 11 cols)

- staff_id [PK] (Int64)
- first_name (string)
- last_name (string)
- address_id (Int64)
- picture (string)
- email (string)
- store_id [PK] (Int64)
- active (Int64)
- username (string)
- password (string)
- last_update (datetime64[ns])

store (fact, 4 cols)

- store_id [PK,FK] (Int64)
- manager_staff_id (Int64)
- address_id (Int64)
- last_update (datetime64[ns])

Internal auto-date tables

LocalDateTable_* / DataTableTemplate_* support Power BI date hierarchies. Listed for completeness; business relationships above do not use them.

DataTableTemplate_a2d2931e...

Full name: DataTableTemplate_a2d2931e-28fc-49d4-8f86-8eba292beccb

Used for: PBI auto-date TEMPLATE (not tied to a business column)

LocalDateTable_2da133f1...

Full name: LocalDateTable_2da133f1-f68d-4b4b-8ac9-f0599639b604

Used for: actor[last_update]

LocalDateTable_4239b3b1...

Full name: LocalDateTable_4239b3b1-3dd6-4664-9851-e18de09a0567

Used for: payment[last_update]

LocalDateTable_50a2d13e...

Full name: LocalDateTable_50a2d13e-fe1d-4c9d-a15d-eb4645f9255d

Used for: country[last_update]

LocalDateTable_5bb36e84...

Full name: LocalDateTable_5bb36e84-426a-46fa-a13a-582ef5d6dc26

Used for: payment[payment_date]

LocalDateTable_6086d83f...

Full name: LocalDateTable_6086d83f-4701-4a3c-bc1d-3559af3b0892

Used for: staff[last_update]

LocalDateTable_6a8a1b8a...

Full name: LocalDateTable_6a8a1b8a-08db-41d5-8644-a433e0a4df0b

Used for: customer[create_date]

LocalDateTable_6f8b0d1f...

Full name: LocalDateTable_6f8b0d1f-465a-4e1d-aeb7-5159a1a4f632

Used for: film_category[last_update]

LocalDateTable_70e91751...

Full name: LocalDateTable_70e91751-9ad0-4624-a9f6-239b895d4c07

Used for: rental[rental_date]

LocalDateTable_87276294...

Full name: LocalDateTable_87276294-e462-4311-ae3b-ed5eff935d96

Used for: address[last_update]

LocalDateTable_874aff1d...

Full name: LocalDateTable_874aff1d-fcfb-4b81-ac6e-ffb89b0ffde4

Used for: language[last_update]

LocalDateTable_8765a386...

Full name: LocalDateTable_8765a386-0fbd-4b49-b6b6-3081c9e155ce

Used for: film[last_update]

LocalDateTable_9aade881...

Full name: LocalDateTable_9aade881-448f-4a31-ad80-a5a3f46b3dae

Used for: store[last_update]

LocalDateTable_9f8673ee...

Full name: LocalDateTable_9f8673ee-bb2e-4fbb-bb1c-ec9378110b48

Used for: customer[last_update]

LocalDateTable_a8f0fb53...

Full name: LocalDateTable_a8f0fb53-b463-4fac-b8e7-a889be094bc7

Used for: rentat[return_date]

LocalDateTable_c928e018...

Full name: LocalDateTable_c928e018-84d6-4ce3-ae3b-63bafef00319

Used for: film_actor[last_update]

LocalDateTable_d0d09aee...

Full name: LocalDateTable_d0d09aee-878f-400a-9b99-c55fb13a7bb3

Used for: rentat[last_update]

LocalDateTable_e54b6b64...

Full name: LocalDateTable_e54b6b64-66e9-4425-bfb5-60e11fa998ee

Used for: city[last_update]

LocalDateTable_ee80cc38...

Full name: LocalDateTable_ee80cc38-2934-4181-972c-f0b7d5752b22

Used for: category[last_update]

LocalDateTable_fc9173f0...

Full name: LocalDateTable_fc9173f0-2a64-4402-a7d0-c24da1027bc5

Used for: inventory[last_update]